



RESEARCH ARTICLE

Load Flow and Short-Circuit Analysis of a Two-Generator Meshed Transmission Network Using ETAP

Arijit Ganguly^{1*}, M.Vamshi Kumar² and Sayan Bhattacharya²

¹Assistant Professor, Department of Electrical Engineering Brainware University, Kolkata, West Bengal, India

²Student, Department of Electrical Engineering, Brainware University, Kolkata, West Bengal, India

Received: 11 Mar 2026

Revised: 10 Apr 2026

Accepted: 09 May 2026

*Address for Correspondence

Arijit Ganguly

Assistant Professor,

Department of Electrical Engineering

Brainware University,

Kolkata, West Bengal, India

E. Mail: er.arijitganguly@gmail.com



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ABSTRACT

This paper presents an exhaustive load-flow and short-circuit analysis of a two-generator, five-bus meshed transmission network utilizing the ETAP simulation platform. The system that was looked at works at the 161 kV transmission level. It has two generating units that are connected to the grid by 15/161 kV step-up transformers. It also has four lumped loads that are spread out over different buses. The Newton-Raphson method is used to find the network's steady-state operating condition. To check how well the system works under normal conditions, we look at the bus voltage magnitudes, line currents, real power flow, and reactive power flow. The load-flow results show that all of the buses have acceptable voltage profiles and that the network is working within its allowed loading limits. In addition, real and reactive power distributions across the transmission lines are observed to be well balanced. The IEC 60909 standard is used to perform short-circuit analysis to determine which fault current levels are present at certain buses. Different types of faults, namely three-phase, line-to-ground, line-to-line, and double-line-to-ground faults, are considered to represent both symmetrical and unsymmetrical fault conditions. The obtained fault currents are used to evaluate the duty of circuit breakers and to assess the suitability of existing protection arrangements. The differences in fault current levels between buses are also looked at to find the most important places in the network where faults happen. The results show that the studied system is electrically stable during steady-state conditions. The calculated fault levels stay within practical limits for protection design. The study confirms that ETAP is a reliable simulation tool for combined load-flow and short-circuit investigations. The outcomes of this work offer a helpful technical foundation for selecting equipment ratings, conducting protection coordination studies, and planning future expansions of interconnected meshed transmission networks.

Keywords: Load Flow Analysis, Short-Circuit Analysis, ETAP, IEC 60909, Meshed Transmission Network, Newton-Raphson Method, Fault Current Calculation.





INTRODUCTION

Load flow studies are crucial for standard electrical system development and functioning [1]. The load flow produces a sinusoidal pattern in the steady state for the whole system. It determines the voltage, active and reactive power, and line losses. Load flows between lines depend primarily on voltage and power angle. Power travels from the generating station to the load via numerous branches in a power system network. Load flow, sometimes called power flow, is the movement of both active and reactive power. One crucial tool that power engineers use to plan and assess a power system's steady-state performance is load flow analysis [2]. As industrialization progressed, the power system and the size of the load flow equation both continued to grow, reaching several thousand. No numerical method can converge to a valid answer with such growth. Power engineers must therefore look for more dependable techniques. The challenge facing the power industry is determining which approach is best for analyzing a power system. To select the optimal approach for power flow analysis, a high degree of accuracy and a faster solution time are necessary.

LITERATURE REVIEW

The authors [3] demonstrated that addressing reactive power is crucial for enhancing system efficiency, reducing energy losses, and stabilizing voltage levels. Simulation platforms like ETAP are instrumental in performing load flow analyses and designing optimal compensator configurations. Nevertheless, there remains a pressing need for additional studies on integrated system-level analysis, especially within laboratory and distribution networks that incorporate renewable energy sources. The authors [4] represent that solar PV integration in distribution networks enhances voltage profiles and stability in the IEEE 33-bus system, but disrupts fault current levels and power flow, undermining conventional relay coordination. While past research has targeted over current and distance relay optimization along with distributed generation placement for reliability, there is a gap in studies addressing voltage stability and coordinated relay operation during both grid-connected and islanded scenarios. This study aims to analyze the stability of a PV-integrated IEEE 33-bus network under these conditions with a focus on coordinated protection and selective load disconnection. The authors [5] represent that over current relay coordination for an IEEE 9-bus power system when photovoltaic (PV) generation is integrated, and compare the protection behavior with and without PV using ETAP software. Load-flow and short-circuit analyses are carried out for different buses and fault locations to determine relay settings and breaker operating sequences under both conventional and PV-integrated conditions. The results show that integration of PV changes fault current levels and relay operating sequences, and relay set-points must be adjusted depending on the percentage of power shared by PV sources. It is observed that, in PV-connected buses, the fault current contribution is lower, so current-transformer (CT) ratios and relay settings need to be reduced compared to the conventional system. The authors [6] represent the electrical network of a 1240 MW combined-cycle power plant using ETAP by performing load-flow, voltage-stability, and short-circuit studies. It identifies weak buses, voltage-stability limitations, and fault-level issues, and then proposes practical remedial measures (mainly reactive power/voltage support and network improvements) to enhance plant reliability and safety. The author [7] presents a fault-arc circuit breaker that detects both arc voltage and short-circuit current to improve arc-extinguishing performance. By using real-time measurements of the arc voltage waveform together with short-circuit current characteristics, the breaker can more accurately determine arc conditions and trigger faster, more reliable interruption. The approach aims to enhance fault detection sensitivity and breaking efficiency, especially under complex fault scenarios, compared to traditional breakers that rely mainly on current detection alone.

Power Flow Analysis

Power system planning and operation are significantly influenced by load flow analysis. This analysis provides essential information about real power flow, reactive power flow, voltage magnitudes, and their corresponding angles at various buses. Understanding the voltage magnitudes and angles is crucial for ensuring they remain within





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acceptable limits for optimal performance [3]. Once we determine the voltage magnitude and angle at each bus in a steady state, we can calculate the real and reactive power flows between the buses. This information is vital for assessing the loading conditions of the transmission lines. Additionally, the power flow analysis helps us identify the losses between each bus. Keeping these losses within acceptable limits is essential for system efficiency. The real and reactive power consumption corresponding to the loads is assumed to remain constant in load flow studies. Additionally, the generator terminal voltages are thought to be strictly controlled and, therefore, remain the same. The main objective of load flow analysis is to determine the voltage magnitude and angle at each bus, given the specified generator outputs and loads. To make this process easier, we classify the different buses in the power system as outlined below [4][3]:

Load Bus: There are no generators attached to this type of bus; both the generated real power (P_G) and reactive power (Q_G) are zero. The load drawn by these buses is characterized by real power (P_L) and reactive power (Q_L), with a negative sign indicating that power is flowing out of the bus. As a result, they are commonly referred to as P-Q buses. The load flow analysis aims to calculate the voltage magnitude ($|V_i|$) and angle (δ_i) at each bus.

Voltage Controlled Bus: These are the buses with generators attached. In such buses, power generation is controlled by a prime mover, while the generator excitation regulates terminal voltage. Constant P_G and $|V_i|$ can be specified for these buses by controlling the input power using turbine governors and the bus voltage through an automatic voltage regulator. This is why these buses are also called P-V buses. The reactive power delivered by the generator Q_G is determined by the system setup and cannot be specified in advance. In addition, we need to determine the unknown angle (δ_i) of the bus voltage.

Slack bus: Typically, this bus is numbered 1 for load flow studies. This bus establishes the angular reference for all other buses. The angle difference between two voltage sources determines the actual and reactive power flow; the angle of the slack bus is unimportant. However, it serves as the standard for measuring the angles of other bus voltages. The bus's angle is typically set at 0° . Additionally, it is expected that the voltage of the bus is known [5]. In this research, we used the Newton-Raphson approach for simulation.

Short Circuit Analysis

A short circuit occurs when an unintended or intentional low-resistance connection is established between two points in an electrical circuit, bypassing a portion of the circuit. When these two points are connected with low resistance or impedance, current will flow through this alternate path. In an electric circuit, the current flows along the path that offers the least resistance. In short-circuit situations, the current level can suddenly increase by hundreds or even thousands of times, making it potentially lethal. Short-circuit current is the term used to describe this high current flow via the alternate path. A short circuit can have catastrophic consequences. The system's capacity to generate short-circuit current and the time duration it is allowed to operate under short-circuit conditions determine the results. Electrical arcs may form locally at the short circuit location, resulting in fire, conductor welding damage, and insulation damage. Electrodynamic forces on the malfunctioning circuit could cause bus bars and cables to distort, and an extreme temperature increase could harm insulation. Short-circuit current is the term used to describe this high current flow via the alternate path [6][8]. A three-phase power network may experience several types of short-circuit issues. These defects are often classified as symmetrical (balanced) faults and unsymmetrical (unbalanced) faults. A symmetrical or three-phase fault affects all three phases of the electricity system. Conversely, an unsymmetrical defect affects one or two phases. Additionally, Thevenin equivalent sequence circuits seen from the fault location in power systems are used to assess all short circuit faults. The types of short-circuit faults are line-to-ground faults, line-to-line faults, double line-to-ground faults, three-phase faults, and three-phase-to-ground faults.

Symmetrical Fault

A symmetrical fault occurs when all three phases are connected or shorted to the ground [7]. The duration of a fault can be divided into three categories:

- The sub-transient phase that takes place right on the fault site. Only a few cycles pass throughout it.
- For tens of cycles, the transient phase occurs.
- The duration of the steady-state phase will be extended.





Unsymmetrical Fault

In an unsymmetrical fault, just one or two phases are affected. Unsymmetrical faults cause the three-phase lines to become unbalanced. Faults of this type occur between lines or between lines and the ground. Asymmetry in the line impedances is an unsymmetrical shunt fault, whereas an imbalance between phases or between phase-to-ground is an unsymmetrical series fault [6][7]. Due to their exposure to external factors, transmission lines experience the bulk of problems. A short circuit might be caused by tree branches, flashover in line insulators could be caused by lightning strikes, tower collapse could be caused by strong winds, and mechanical failure of the line or insulator could be caused by ice loading and wind. Cable, circuit breaker, generator, motor, and transformer faults are much less common. Relay settings and coordination, as well as choosing the appropriate circuit breaker rating, require fault analysis. While symmetrical components are used to evaluate unsymmetrical faults, symmetrical faults are examined on a per-phase basis. Additionally, short-circuit investigations benefit greatly from the ZBUS matrix.

The following categories apply to shunt faults in a three-phase system:

- **Line-to-Ground (L-G) fault:** One phase is assigned to ground in a single line-to-ground fault; during this phase, the voltage and current are both zero. In power systems, this is the most typical form of fault. Approximately 70% to 80% of all faults are caused by it. It usually happens as a result of lightning or contact between a conductor and a grounded building [7][8].
- **Line-to-Line (L-L) fault:** High winds can cause a line-to-line fault, when one phase may come into contact with another phase. Line-to-line faults comprise approximately 8% to 15% of transmission line faults. It happens when a tree branch falls on top of two power lines or when a large bird stands on one transmission line and touches the other [7][8].
- **Double Line-to-Ground fault:** The voltage value of that phase is zero, and the current value is large in a double line-to-ground fault. This type of fault can occur when a tree falls, and the two phases come into contact with the ground. Additionally, in line-to-ground fault circumstances, two phases rather than one will be involved. This kind of fault constitutes approximately 10% of all transmission line faults. The distribution system is where open circuit and short circuit faults occur, while the generator's terminal is where symmetrical and unsymmetrical faults mostly occur [6][7].
- **Three-phase Fault:** These faults are short-circuited faults that can result from physical accidents, insulation breakdown during lightning strikes, or other natural occurrences like wind and icfall. This short circuit current can harm the system's power system equipment of the system due to it being 6% to 10% more than the typical operating current. This kind of issue is actually rare, as evidenced by the fact that it accounts for only 5% of all transmission line faults [7].

IEC 60909-0 short-circuit current parameters:

Initial Symmetrical Short-Circuit Current (I_k''): When a three-phase fault takes place in such an instance of system, the maximum initial short-circuit current will occur. This current is the short-circuit current's root mean square value of its original alternating current component [8]. It is employed in the computation of maximum short-circuit current. The rms value of the AC symmetrical component of an accessible short-circuit current is relevant at the instant of short-circuit if the impedance persists at zero time.

$$I_k'' = \frac{c V_n}{\sqrt{3} Z_k}$$

where, Z_k is the equivalent impedance at the precise location of the failure, V_n is the normal voltage.

Peak Short-Circuit Current (I_p): It is the maximum instantaneous value of the available short-circuit current [8].

$$I_p = \sqrt{2} k I_k''$$

where,

$$k = 1.02 + 0.89 e^{-3 \frac{R}{X}}$$

The ratio R/X of determines the magnitude of k at the fault location of the system.

Symmetrical Short-Circuit Breaking Current (I_b): It is the root mean square value of an integral circle of the symmetrical ac components of the accessible short circuit current at the moment of contact separation of a switching device's initial pole [7][8].





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The initial symmetrical short-circuit current and the symmetrical short-circuit breaking current are equivalent when the fault develops far from the generator.

$$I_b = I_k''$$

When the fault occurs near the generator, by combining contributions from individual machines to obtain the symmetrical short-circuit breaking current.

For a synchronous machine,

$$I_b = \sigma I_k''$$

For an asynchronous machine,

$$I_b = \sigma q I_k''$$

where, σ and q are the factors of AC decay.

Steady State Short-Circuit Current (I_k): Compared to the initial short-circuit current, the steady-state current calculation will yield a less accurate result. Assume that synchronous machines operate at maximum excitation when the highest steady-state short-circuit current can be calculated [7][8]. The appropriate equivalent impedance for the minimal steady-state current is determined using the steady-state short-circuit current value for the synchronous machines. For these kinds of computations, all asynchronous motors are disregarded. It is the short-circuit current's root mean square gain that remains after the transitory event has diminished.

$$I_{k(max)} = \mu_{max} \cdot I_g$$

$$I_{k(min)} = \mu_{min} \cdot I_g$$

where, I_g is the rated current of the generator, μ is depends on the excitation voltage of the generator, the ratio between its initial symmetrical short-circuit current and rated current, and other parameters of the generator. When the synchronous generator is operating at its maximum excitation voltage, the maximum steady-state short-circuit current $I_{k(max)}$ is calculated. When the synchronous generator is operating at the lowest excitation voltage, the minimum steady-state short-circuit current $I_{k(min)}$ is calculated.

Voltage Factor (c): In order to estimate the lowest and highest current, it is the section that modifies the value of the same voltage source. Internal impedances are a representation of every machine [8]. There are various methods for correcting system voltages and transformer impedance in the event of an off-nominal tap setting. There are two possible settings for the transformer tap: the nominal position and the running position.

Network Description

This is a two-generator, five-bus (161 kV) meshed transmission system, with generators connected through 15/161 kV step-up transformers and four lumped loads distributed on the transmission buses.

METHODOLOGY

By completing data from a one-line diagram in the form of generator, load, cable, bus, transformer, and numerous other supporting components, the IEEE 5-bus electric power system network conducts this power flow analysis. Several steps are included in this power flow study, including data gathering, one-line diagram creation, simulations, and outcomes. The flowchart figure below illustrates the power flow study method.

RESULT

Load- Flow Result: Table. 1

From the Fig. 3, it can be observed that the voltage loading slightly decreases as the bus number increases. At Bus1, the voltage loading is the highest at approximately 94.39%. As the power flows through the network, the voltage level gradually drops due to line impedance and power losses. At Bus2, the voltage loading decreases to around 94.33%. The value at Bus3 remains almost the same, indicating a stable voltage profile between these buses. Moving further, Bus4 and Bus5 show a small additional decrease, reaching approximately 94.31% at Bus5. The overall



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variation in voltage loading is very small (about 0.08%), which indicates that the system voltage profile is stable and well-regulated across all buses. There are no significant voltage drops, suggesting that the power system is operating within acceptable voltage limits.

Short-Circuit Analysis

From the table, the 3-phase fault produces the highest fault current at all buses. This occurs because a 3-phase fault is a balanced fault with the lowest equivalent impedance, allowing maximum current to flow. The Line-to-Ground fault currents are slightly lower, around 7.46 kA. These faults are very common in real power systems but produce less current than symmetrical faults because of additional impedance in the ground path. The Line-to-Line faults show the lowest current values, approximately 6.98 kA, because only two phases are involved, and the fault impedance is higher. The Line-to-Line-to-Ground faults produce currents around 7.87 kA, which are higher than L-L faults but slightly lower than the 3-phase fault. The fault current values for Bus2, Bus3, Bus4, and Bus5 are very close to each other. The difference is extremely small (around 0.003 kA), which indicates that the network impedance between these buses and the power source is almost identical. The peak current (i_p) values are approximately 2.5–2.6 times the initial symmetrical current (I_k). This high peak current occurs due to the DC offset component in the first cycle of the fault. Based on the analysis, the maximum fault current in the system is approximately 8.035 kA. Circuit breakers and protection equipment must be rated higher than this value to safely interrupt faults. The peak current (approximately 21 kA) is important for determining the mechanical strength of switchgear and bus bars.

Transient Analysis

Case – 1: When a short-circuit fault occur effect for alternator G1

From Fig. 4 at the start ($t \approx 0$ s), the angle is near zero electrical degrees, indicating the generator is synchronized with the reference. Then the angle increases smoothly, showing the rotor is accelerating relative to the reference. Around the middle of the plot, the angle reaches about +160 electrical degrees to +170 electrical degrees. Suddenly, the angle drops to about -170 electrical degrees. This condition is angle wrapping, not a physical jump. When the rotor passes +180 electrical degrees, it is represented as -180 electrical degrees in relative angle plots. The same pattern repeats gradually increase $\rightarrow$ near +180 electrical degrees $\rightarrow$ wrap to -180 electrical degrees $\rightarrow$ increase again. The generator rotor is continuously advancing relative to the reference. The wrapping indicates multiple electrical revolutions. This behavior often occurs during loss of synchronism or large rotor oscillations in transient stability simulations.

From Fig. 5, the angle begins close to zero electrical degrees. Over time, it continues to increase. The accelerating, nonlinear curve reaches about 460 electrical degrees to 480 electrical degrees in one second. The rotor is always moving ahead. The angle continues to increase because the graph does not wrap at ± 180 electrical degrees. The accelerating slope shows that the generator is accelerating, most likely as a result of (i) the mechanical input power and electrical output power are not balanced, or (ii) a disturbance. According to the plot, Generator 1 is accelerating over time, most likely as a result of either (i) an increase in mechanical input power or (ii) a dynamic response after a control action or disturbance in a power system simulation. A steady, controlled speed increase void of oscillations is indicated by the smooth linear ascent. The current quickly increases to about 7500 A and 7800 A. This denotes a temporary state, like the occurrence of a fault, unexpected load connection, or the system's switching event. For a brief period, the current is quite high and slopes somewhat downhill. After that, the current suddenly decreases to between 1100 A to 1200 A. The system has attained a new stable operating condition when the current becomes almost constant following the dip.

Case – 2: When a short-circuit fault occur effect for alternator G2

The power angle (δ) represents the angular displacement between the rotor magnetic field of a synchronous generator and the stator rotating magnetic field. The steady decline suggests that Generator 2 is either speeding in relation to the synchronizing field or losing synchronism with the system reference. Such actions could be brought on by: (i) a significant disruption or malfunction in the system; (ii) inadequate synchronizing torque; and (iii) an abrupt disparity between mechanical input power and electrical output power. According to the Fig. 8, the rotor angle of Generator 2 appears to be continuously changing over time, which could be a sign of dynamic instability or loss of

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synchronism if the angle continues to grow. The starting speed of Generator 2 is roughly 1500 rpm, which is the standard synchronous speed for a generator operating in a 50 Hz power grid. The generator is initially running under stable steady-state circumstances because the speed stays almost constant for a brief period of time. The speed gradually starts to drop after the first moment. Over time, the speed continues to drop steadily and nonlinearly, indicating that the generator is slowing down. The speed decreases to about 1460 rpm –1465 rpm by the conclusion of the duration. At first, the generator current stays extremely low, near 0–500 A, suggesting that the system is operating normally or under light load. The current quickly rises, first to about 2000 A and then quickly to about 18,000 A –19,000 A. A significant transient current in the system is shown by the current reaching a maximum value close to 19,000 A. The stream gradually drops over time after reaching the apex. The system is settling following the brief disruption when the current drops to roughly 10,000 A –11,000 A by the conclusion of the time interval. A brief disruption in the power system is indicated by the significant current spike, such as: (a) A malfunctioning short circuit, (b) an unexpectedly high load connection, (c) an operation of system switching. The consistent decrease in current indicates that the generator current is returning to a new steady-state value and that the system protection or control mechanism is operating.

CONCLUSION

This study presented a detailed power-flow and short-circuit analysis of a two-generator, five-bus meshed transmission network using the ETAP simulation platform. The Newton-Raphson method for the load-flow studies results show that all buses function within adequate voltage limits, and the real power and reactive power loadings remain within permissible ranges. Short-circuit analysis based on IEC 60909 indicates that three-phase and unsymmetrical fault currents vary across buses but remain within the interrupting capability required for practical protection design. The obtained fault levels provide a reliable basis for selecting circuit-breaker ratings and protection settings. Overall, the results confirm that the studied network is electrically stable under normal operating conditions and that the adopted analysis approach is suitable for planning, protection coordination, and future system expansion studies.

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Table 1 Load Flow Result

Bus Id	Percentage of Voltage Loading	Current in ampere	Active Power Loading in MW	Reactive Power Loading in MVar
Bus1	94.39	404.1	90.545	55.81
Bus2	94.33	259.3	57.964	35.969
Bus3	94.33	340.9	76.312	47.092
Bus4	94.32	113.4	25.379	15.679
Bus5	94.31	111.5	24.936	15.454

Table 2 Short Circuit Analysis

Bus	3-Phase Fault			Line-to-Ground Fault				Line-to-Line Fault				Line-to-Line-to-Ground Fault			
	"	"	"	"	"	I_b	I_k	I''_k	i_p	I_b	I_k	I''_k	i_p	I_b	I_k
Bus 2	8.03 4 kA	20.88 4 kA	4.96 4 kA	7.46 2 kA	19.39 5 kA	7.46 2 kA	7.46 2 kA	6.98 3 kA	18.15 0 kA	6.98 3 kA	6.98 3 kA	7.87 0 kA	20.45 7 kA	7.87 0 kA	7.87 0 kA
Bus 3	8.03 5 kA	20.91 0 kA	4.96 5 kA	7.46 2 kA	19.42 0 kA	7.46 2 kA	7.46 2 kA	6.98 3 kA	18.17 3 kA	6.98 3 kA	6.98 3 kA	7.87 3 kA	20.48 9 kA	7.87 3 kA	7.87 3 kA
Bus 4	8.03 3 kA	20.75 2 kA	4.96 4 kA	7.46 0 kA	19.27 3 kA	7.46 0 kA	7.46 0 kA	6.98 1 kA	18.03 5 kA	6.98 1 kA	6.98 1 kA	7.86 9 kA	20.32 9 kA	7.86 9 kA	7.86 9 kA
Bus 5	8.03 2 kA	20.74 6 kA	4.96 4 kA	7.46 0 kA	19.26 8 kA	7.46 0 kA	7.46 0 kA	6.98 1 kA	18.03 0 kA	6.98 1 kA	6.98 1 kA	7.86 8 kA	20.32 1 kA	7.86 8 kA	7.86 8 kA

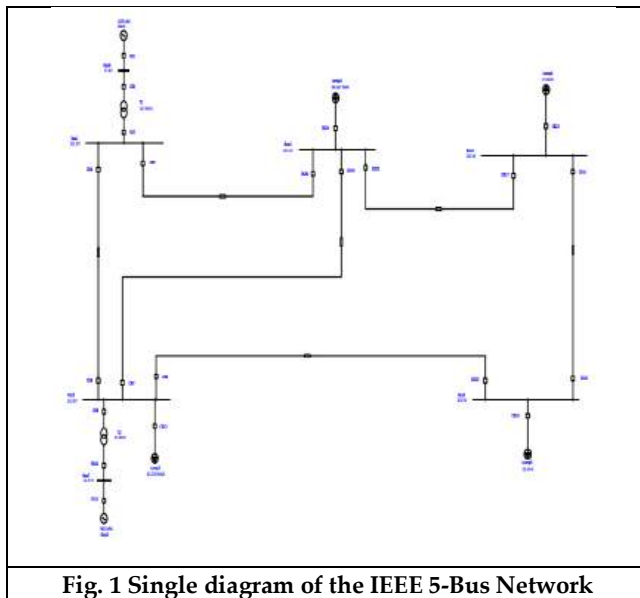


Fig. 1 Single diagram of the IEEE 5-Bus Network

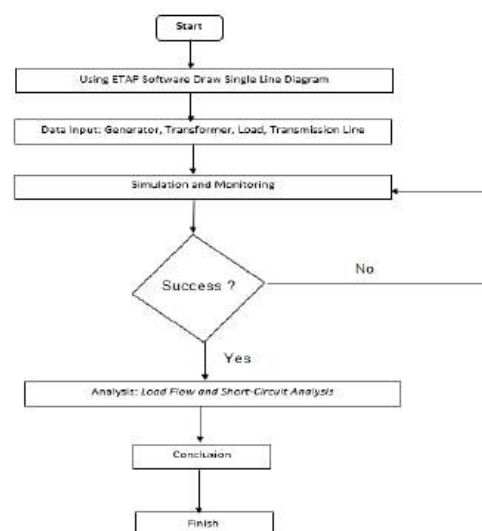


Fig. 2 Flow Chart





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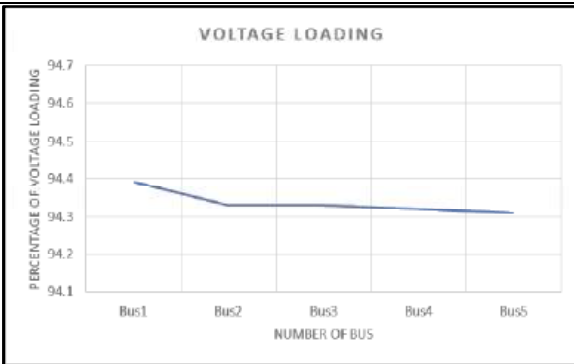


Fig. 3 shows the percentage of voltage loading at different buses (Bus1 to Bus5) in a power system network.

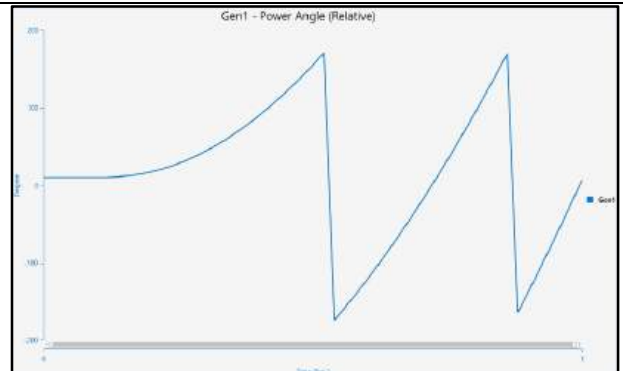


Fig. 4 Gen.1 Power Angle (Relative)

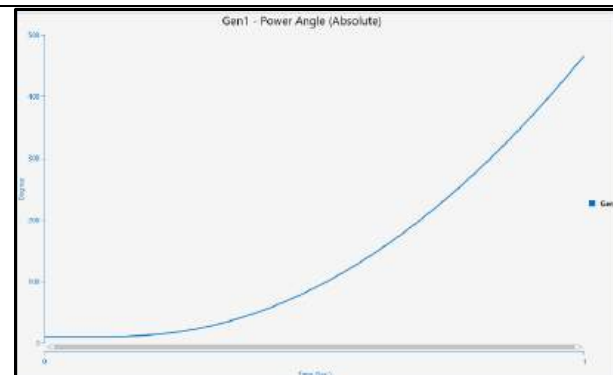


Fig. 5 Gen.1 Power Angle (Absolute)

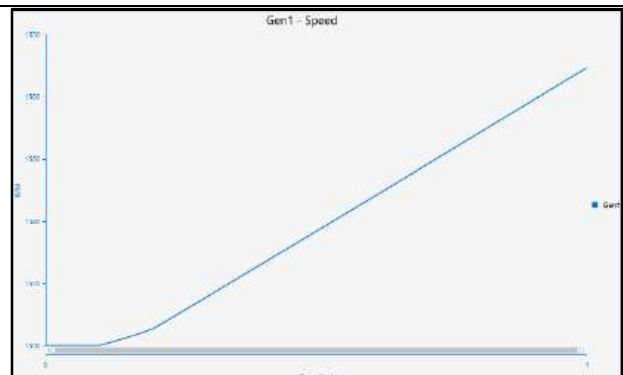


Fig. 6 Gen1- Speed

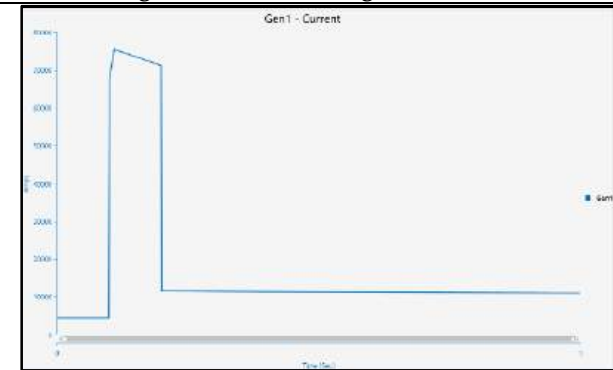


Fig. 7 Gen1 - Current

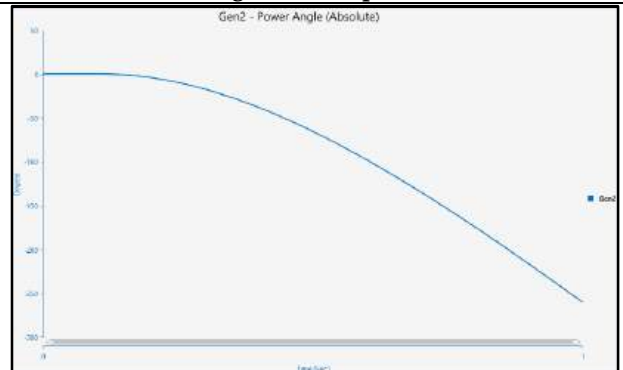
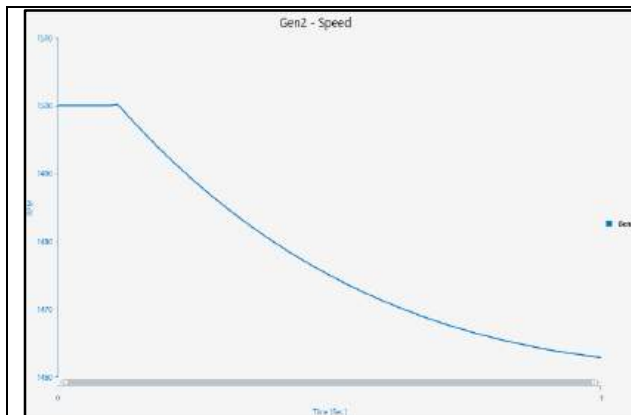
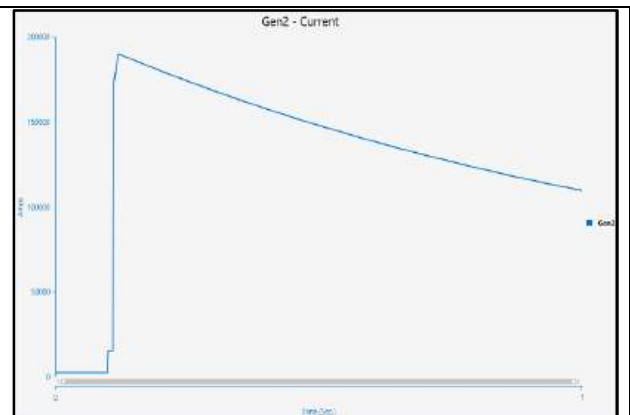


Fig. 8 Gen2 Power Angle (Absolute)



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